

Paula Brüdrich

+49 1575 1334654 | brudricp@tcd.ie | [linkedin.com/in/paula](https://www.linkedin.com/in/paula) | Dublin, Ireland & Bielefeld, Germany

PROFILE SUMMARY

Electronic & Computer Engineering student and Foundation Scholar at Trinity College Dublin, with **first-place academic awards in each of the first three years** of engineering study. Hands-on experience in PCB design, embedded systems and robotics through drone hardware development, vision-integrated robotics and a sensor-driven autonomous buggy. Seeking technical roles in electronics, embedded systems and robotics.

TECHNICAL SKILLS AND LANGUAGES

- Electronics & Embedded: **PCB** design & layout, **signal-integrity** analysis, microcontrollers, sensor interfacing, hardware prototyping, soldering, I2C, SPI, PWM, power-supply debugging, Verilog (**Vivado**)
- Programming & Robotics: C, **C++**, PID control, stereo vision, hand-eye calibration, WiFi/UDP communication, basic ARM Assembly, MATLAB for coursework
- Engineering Tools: **Autodesk Fusion Electronics**, **SolidWorks**, ANSYS Acoustics, 3D printing
- Languages: German: Native | English: C2

EXPERIENCE

Patch Summer Accelerator - Vision-Integrated Robotics Team

July 2026 – Aug. 2026

Dogpatch Labs

Dublin, Ireland

- Integrated an **rc-visard stereo camera** with a **BCN3D-Moveo** robotic arm, implemented hand-eye **calibration** and handled data transfer to a **Python** environment for processing
- Worked with 3 teammates to complete the project in **7 weeks** and organized **6000€+** worth of equipment through collaborations using only **170€** of our hardware **budget**
- Performed a live demo and pitch in front of **150+ audience members** from the Irish start-up community

Lead Powertrain Engineer

Aug. 2025 – July 2026

Formula Trinity

Dublin, Ireland

- Responsible for the **entire powertrain** including engine and exhaust, intake, cooling, fuel and lubrication systems
- Designed Formula Trinity's first custom, **lab-tested muffler** and supervised the development of the team's first **carbon-fibre intake** plenum
- Recruited and trained **all 8 new members** and represented the team at FSUK in **scrutineering, design** and **cost** judging, successfully proving rule-compliance across all parts

Hardware Development Intern for Drones in Logistics Applications

May 2025 – July 2025

HHLA Sky, formerly Third Element Aviation

Bielefeld, Germany

- Contributed to a full design cycle of a **6-layer, 400-component** Flight Supply Board integrating **5 microcontrollers** and their functions onto a single PCB
- Performed **PCB layout** and **signal-integrity analysis** using Autodesk Fusion Electronics
- Assembled and soldered the first **5 prototypes** and supported load testing of the boards
- Disassembled, updated and reassembled a test copter within a week to replace outdated components

Junior Powertrain Engineer

Oct. 2024 – Aug. 2025

Formula Trinity

Dublin, Ireland

- Represented the department in front of design judges and achieved a score of 25/35, more than **3 times higher** than the team's **previous result**, with personal endorsement from the judges
- Responsible for the design of the air intake system to optimize for **fluid behavior** and **resonance effects**
- Designed the air-intake system in **SolidWorks** and sized plenum volume and runner lengths using resonance models including Helmholtz resonators

EDUCATION

Trinity College Dublin

Dublin, Ireland

MAI Electronic and Computer Engineering, Foundation Scholar, expected graduation May 2028 Sep. 2023 – Present

- Received the title of **Scholar** through voluntary exams covering the first 3 semesters of the course
- **Ranked first** in Engineering in Years 1 and 2 (91%, 87%) and joint first in Electronic & Computer Engineering in Year 3 (88%).

PROJECTS

Sobriety Suite | *Embedded Systems, C++/Arduino, I2C, SPI, SolidWorks*

Jan.–Apr. 2026

- Developed an **Arduino UNO R4 WiFi embedded system** integrating an MQ3 analog alcohol sensor, MAX30102 heart-rate sensor over **I2C**, 128x64 OLED over **SPI**, user controls and LM386-based audio feedback
- **Diagnosed** power-supply issues causing significant audio noise and redesigned the **power architecture** with separate sensor and audio supplies, eliminating speaker hum and improving sensor stability
- Designed and **3D-printed the complete enclosure and sensor mounts in SolidWorks**, including custom breathalyser and heart-rate fixtures to create a compact, portable device and improve measurement repeatability

Autonomous Buggy | *C++/Arduino, PID Control, Sensors, WiFi/UDP*

Jan.-Apr. 2025

- Built an **Arduino-controlled autonomous mobile robot** integrating DC motors, wheel encoders, IR line sensors, ultrasonic obstacle detection and a HuskyLens vision sensor for autonomous navigation
- Implemented and tuned custom **PID controllers** for wheel-speed regulation and 15 cm object following, using encoder RPM and ultrasonic distance feedback & experimentally derived an **RPM-to-PWM calibration curve** to improve low-speed response
- Implemented **vision-based navigation** using AprilTags and a HuskyLens camera and added WiFi communication, a Processing control GUI and **UDP smartphone steering** using phone orientation data

AWARDS AND SCHOLARSHIPS

Stanford-Smith Prize and B.K.P. Scaife Prize in Electronic/Computer Engineering (shared) May 2026

- Awarded to the highest scoring student in the **third year** Electronic/Computer Engineering examinations

Trinity Foundation Scholarship in Engineering

May 2025

- Achieved first honors in voluntary scholarship exams to be awarded the title of Scholar (Sch.)
- Highest average (85%) on the Engineering Scholarship exams in 2025

Anita Newell Scholarship and Francis Spring Prize

May 2025

- Awarded to the highest scoring student in the **second year** Engineering examinations (respectively for female students and regardless of gender)

Anita Newell Scholarship and Marmaduke Backhouse Prize

May 2024

- Awarded to the highest scoring student in the **first year** Engineering examinations (respectively for female students and regardless of gender)